

Example dataset analysis with *OrthoGather*

1. Dataset used:

The example dataset was derived from *Giddey et al.* (2017). A temporal proteome dynamics study reveals the molecular basis of induced phenotypic resistance in *Mycobacterium smegmatis* at sub-lethal rifampicin concentrations. *Scientific Reports*, 7:43858; PMID: 28262820.

Protein identifiers were obtained from the Supplementary materials, which report quantitative proteomic profiles of *M. smegmatis* (*Msm*) across three time points (30, 255 and 300 min) after exposure to sub-lethal rifampicin concentrations. Protein lists from all time points were pooled to create a dataset representing the overall response. They were standardized to UniProt identifiers and are provided in the *S2_Example_dataset_Foreground_Background.xlsx*. Two subsets were extracted:

- *Foreground* (596 proteins): Proteins reported as differentially expressed at any of the time points.
- *Background* (3,244 proteins): The complete set of proteins detected in the study by LC-MS/MS and MaxQuant processing with *M. smegmatis* ATCC 700084 as a reference proteome.

This data was used to illustrate the complete analysis pipeline implemented in *OrthoGather* and was run using the *New Analysis* data input option (Fig. S1).

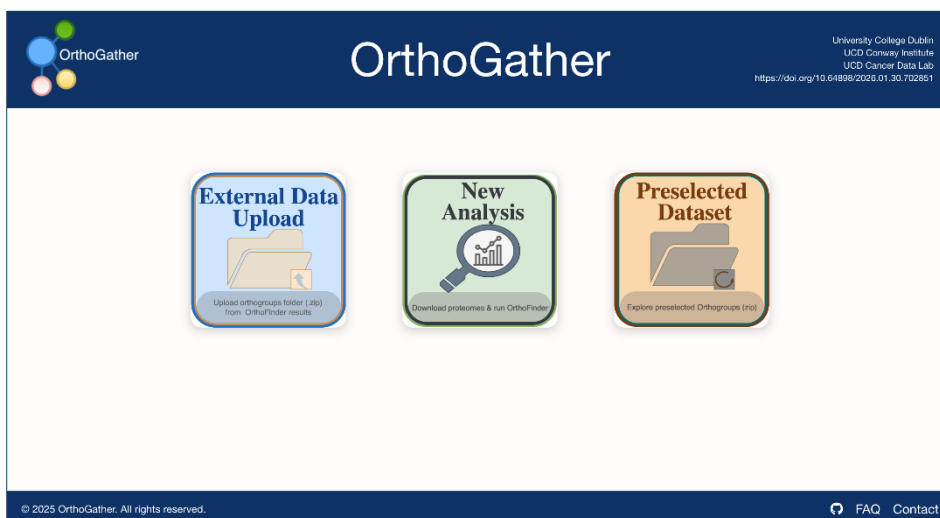


Figure S1. Data input options in OrthoGather

We selected using a subset of six representative bacterial species: *Acinetobacter baumannii* ATCC 19606, *Burkholderia cenocepacia* J2315, *Escherichia coli* K-12, *Mycobacteroides abscessus* ATCC 19977; *Mycobacterium smegmatis* ATCC 700084; *Pseudomonas aeruginosa* PAO1 (Fig. S2). *Msm* is a rapidly growing non-tuberculous mycobacterium (NTM) widely used as a model organism in studies of mycobacterial disease and antibiotic resistance. Although not considered pathogenic, rare case reports describe opportunistic infections, primarily in immunocompromised individuals. Its genome remains relatively poorly annotated, with many functional assignments inferred from other *Mycobacterium* spp. *M. abscessus*, a closely related NTM, is in contrast well recognised as an opportunistic pathogen and is

increasingly viewed as transitioning towards a true pathogen. It is poorly annotated as well. *B. cenocepacia*, *M. abscessus*, and *P. aeruginosa* represent typical CF pathogens, with *A. baumannii* also occasionally implicated in CF lung infections. The *E. coli* K-12 strain was included to support downstream functional annotation steps due to its high-quality genome and extensive annotation (>75%)

The selection of these specific species allows the user to assess whether proteins showing altered abundance in *Msm* following exposure to sub-lethal concentrations of rifampicin are species-specific or conserved in pathogenic mycobacteria, other CF-associated antibiotic resistant pathogens, supporting hypothesis generation regarding potential rifampicin responses in those organisms.

Figure S2. Initial species selection option in the *New Analysis* data input option in *OrthoGather*.

2. OrthoFinder outputs:

After downloading the proteomes, OrthoFinder was executed automatically, generating an output folder named *Orthogroups*. The four key files form the structural basis for the visualizations and downstream analyses (Table S.2, *S4. Orthogather_outputs.zip*)

Table S2. OrthoFinder output files

File	Description
<i>Orthogroups.tsv</i>	Lists all orthogroups and their corresponding UniProt identifiers for each species.
<i>Orthogroups.GeneCount.tsv</i>	Provides the number of genes (proteins) from each species within every orthogroup.
<i>Orthogroups_UnassignedGenes.tsv</i>	Contains genes that were not assigned to any orthogroup.
<i>Orthogroups.txt</i>	A compact text representation summarizing all orthogroup compositions.

3. Comparative Orthogroup Analysis

The *Comparative Orthogroup Analysis* module allows to assess the overall distribution of shared and unique orthogroups and proteins across the species selected by the user. It was executed using all six bacterial species selected in the previous step (Fig. S3).

COMPLETE ANALYSIS

Below are two graphs that represent the analysis of the 6 species present in the ZIP file you uploaded.

The first graph (Figure 1) shows the distribution of proteins by orthogroup, while the second graph (Figure 2) illustrates the number of species sharing orthogroups.

Figure 1

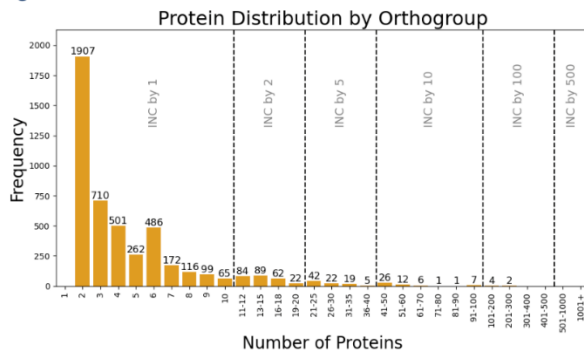
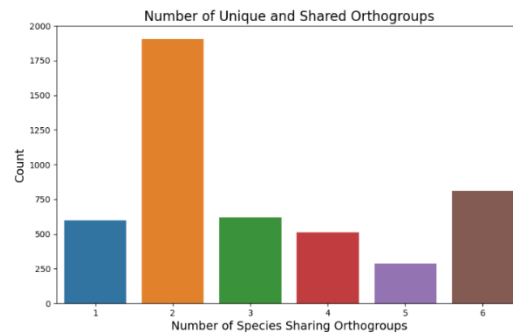
[Download Figure 1](#)

Figure 2

[Download Figure 2](#)

Species Analysis

Select the species you want to analyze

- ☒ Acinetobacter_baumannii_ ☒ Burkholderia_cenocepacia_
☒ Escherichia_coli_ ☒ Mycobacteroides_abscessus_
☒ Mycolicobacterium_smegmatis_ ☒ Pseudomonas_aeruginosa_

Generate Figures

Figure S3. Comparative orthogroup analysis in OrthoGather. Figures are generated to show the general protein distribution within orthogroups and information about number of species sharing orthogroups (top). The user can further choose the species (bottom) used to generate UpSet Plots .

This step generated two global visual summaries illustrating the distribution of orthogroups (Fig. S4) and proteins (Fig. S5). The complete underlying data for these visualizations are provided in *Data_UpsetPlot_All.xlsx* in the *S3. Orthogather_outputs.zip* file. This Excel file contains detailed protein composition for each orthogroup corresponding to the species intersections shown in the UpSet plots. Each worksheet represents a specific intersection combination corresponding to a unique subset of species that share orthogroups and proteins. Within each worksheet, rows correspond to orthogroups, while columns are defined as: *Orthogroup ID*: Identifier of the orthogroup as reported by OrthoFinder. Unique for each separate analysis *Subsequent columns*: UniProt accession codes of the proteins from each species present in the orthogroup. *Empty cells*: Indicate that the corresponding species does not contribute proteins to that orthogroup in the given intersection.

The UpSet Plots generated for this example dataset show that 91 orthogroups encompassing 404 proteins are unique to *Msm*, whereas 810 orthogroups comprising 10,988 proteins are shared by all species. It is important to note that one orthogroup may contain multiple proteins from a species or not contain any for a given species.

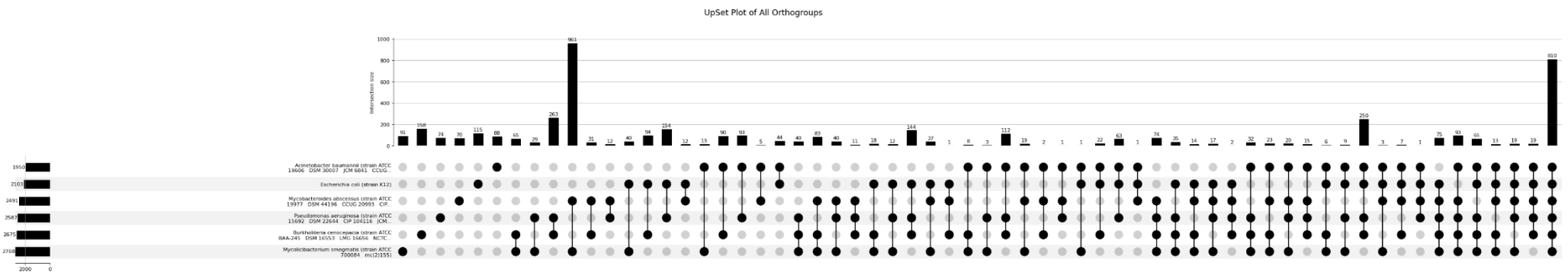


Figure S4. UpSet Plot - all orthogroups. Distribution of orthogroups inferred by OrthoFinder across the selected species. Each vertical bar represents the number of orthogroups shared by a specific combination of species, while horizontal bars indicate the total number of orthogroups per species. High quality figure is available in S3. *Orthogather_outputs.zip* file (*UpSet_Plot_All_Orthogroups.png*)

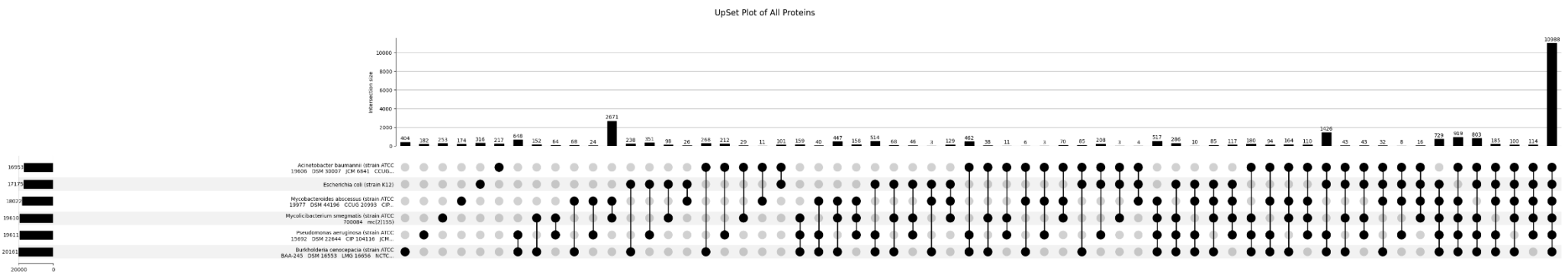


Figure S5. UpSet Plot – all proteins. Distribution of proteins inferred by OrthoFinder across the selected species. This figure represents the same intersections as above and quantifies the total number of proteins contributing to each intersection, allowing comparison of proteome overlap and relative proteome sizes across species. High quality figure is available in S3. *Orthogather_outputs.zip* file (*UpSet_Plot_All_Proteins.png*)

4. Filtered Orthogroup Analysis

This module allows to filter the orthogroups with a user defined protein identifier list (Fig. S6). We have used the 596 *Msm* proteins mentioned above.

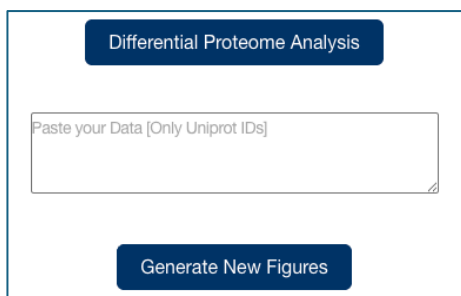


Figure S6. Defining a protein subset in the filtered orthogroup analysis in *OrthoGather*

These proteins were mapped onto the existing orthogroups obtained from the global analysis, and another two upset plots were generated in this step, specifically reflecting the distribution of defined proteins and their orthologs across the six bacterial species. The first visualises orthogroup distribution (Fig. S7), and the second one visualises protein distribution (Fig. S8). Similarly as in the previous step, an accompanying Excel file is generated, containing the detailed corresponding data (S3. *Orthogather_outputs.zip* file, *Data_UpsetPlot_Filtered.xlsx*).

The two UpSet plots generated in this step reveal three orthogroups unique to *Msm*, comprising 12 proteins in total. In the Excel data sheet, a corresponding tab lists this information, including the protein IDs (Table S3). Among the 12 proteins in the three *Msm*-specific orthogroups, four belong to the defined *Msm* set: A0QPS1 and A0QSP2 in orthogroup 1601; A0R771 in orthogroup 2092; and A0R130 in orthogroup 2790. As these proteins are unique to *Msm*, they may represent species-specific factors involved in the response to antibiotic stress. Based on UniProt annotations, A0QPS1 and A0QSP2 are subunits of glycerol dehydratase. Orthogroup 1601 also contains A0QSP1 and A0R5V1, also annotated as glycerol dehydratase subunits, and A0QSP3, annotated as propanediol dehydratase. In Orthogroup 2092, A0R771 (defined by us) is an ABC transporter ATP-binding protein. The other three proteins in this orthogroup are also ABC transporter ATP-binding proteins, with A0R505 and A0R181 annotated as sugar ABC transporter components, allowing us to hypothesise a potential role for A0R771 in sugar transport. Finally, in orthogroup 2790, all proteins remain uncharacterised.

Table S3. Filtered orthogroups belonging unique to *Msm*, as found in the *Data_UpsetPlot_Filtered.xlsx*. Bold: proteins defined by us (foreground, *S2_Example_dataset_Foreground_Background.xlsx*).

Orthogroup ID	<i>Mycobacterium smegmatis_ (strain_ATCC_700084__mc(2)155)</i>
1601	tr A0QPS1 A0QPS1_MYCS2 , tr A0QSP1 A0QSP1_MYCS2, tr A0QSP2 A0QSP2_MYCS2 , tr A0QSP3 A0QSP3_MYCS2, tr A0R5V1 A0R5V1_MYCS2
2092	tr A0QS70 A0QS70_MYCS2, tr A0R181 A0R181_MYCS2, tr A0R505 A0R505_MYCS2, tr A0R771 A0R771_MYCS2
2790	tr A0QNK0 A0QNK0_MYCS2, tr A0QW60 A0QW60_MYCS2, tr A0R130 A0R130_MYCS2

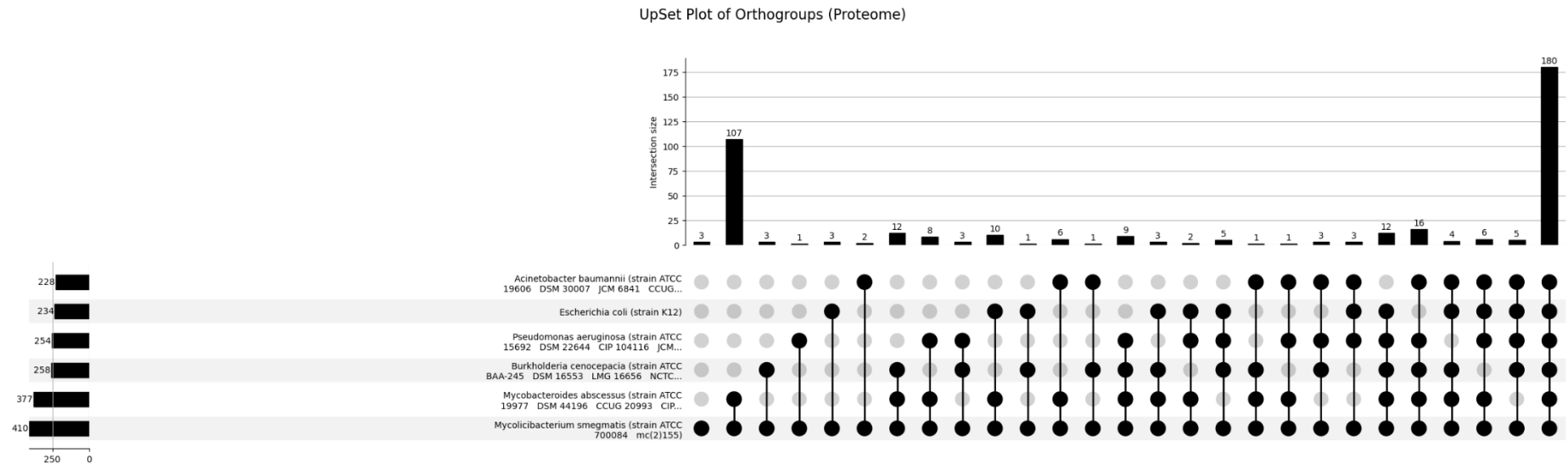


Figure S7. UpSet Plot - filtered orthogroups. Distribution of orthogroups containing at least one of the defined 596 *Msm* proteins. Vertical bars: number of orthogroups in a species subset; horizontal bars: total number of associated orthogroups per species. High quality figure is available in *S3. Orthogather_outputs.zip* file (*UpSet_Plot_Filtered_Orthogroups.png*)

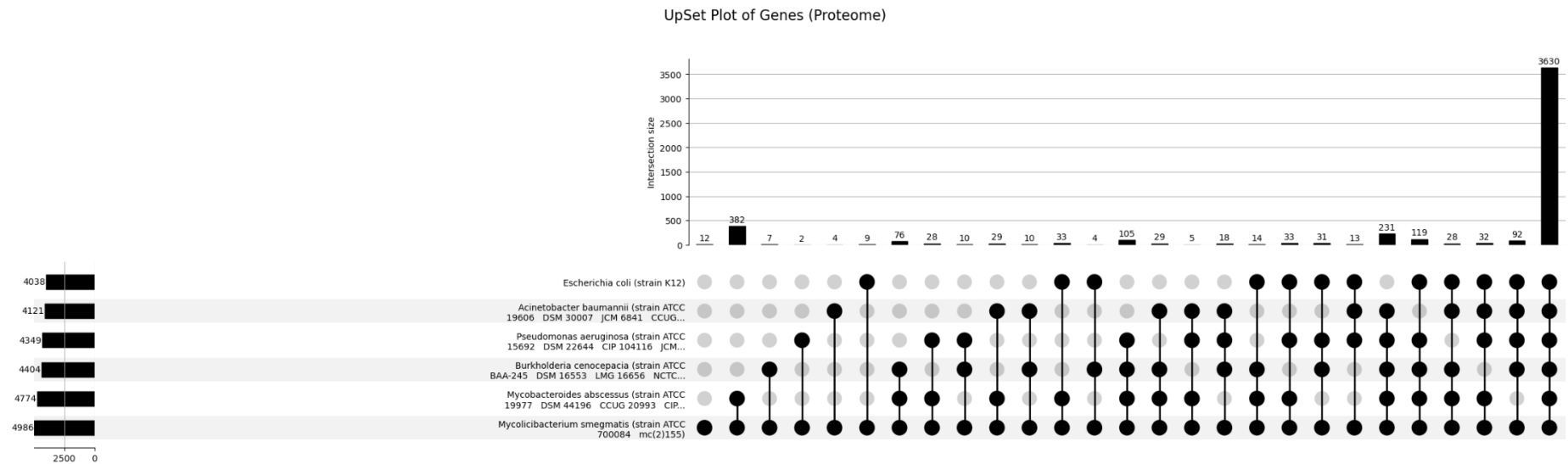


Figure S8. UpSet Plot - filtered proteins. Distribution of proteins in the orthogroups in Fig. S3. Vertical bars: number of orthogroups in a species subset; horizontal bars: total number of associated orthogroups per species. High quality figure is available in *S3. Orthogather_outputs.zip* file (*UpSet_Plot_Filtered_Proteins.png*)

5. Gene Ontology Enrichment Analysis

A separate Gene Ontology module in OrthoGather enables functional interpretation of inferred orthogroups, both before and after filtering, provided that the analysed species have corresponding annotations available in the Gene Ontology Annotation (GOA) repository. This module supports GO enrichment analysis to identify functional categories that are overrepresented within user-defined protein sets, allowing biological processes, molecular functions, or cellular components associated with specific experimental conditions to be highlighted. It is summarised in an Excel file (S3. *OrthoGather Outputs, Gene_Ontology_Analysis.xlsx*; Table S4).

Table S4. Summary of Go term analysis. Worksheets document the specific steps of the GO analysis process (*Gene_Ontology_Analysis.xlsx*).

Sheet name	Description
Meta	General analysis metadata: total number of orthogroups (<i>n_orthogroups</i>), number of species columns (<i>n_species_cols</i>), annotation coverage metrics (<i>rows_nonzero_annotation</i> and <i>rows_zero_annotation</i>).
Initial Groups	List of orthogroups obtained directly from <i>OrthoFinder</i> before any filtering or annotation-based selection.
Filtered Groups	All orthogroups that contain at least one protein belonging to a species with an available GOA file.
Removed Groups	Orthogroups containing no proteins from species with an available GOA file. Excluded from downstream GO analyses due to the absence of GOA resources
Groups of Interest	Refined subset of the filtered orthogroups. Only proteins from annotated species are kept.
Species & GOA Map	Mapping between each analysed species and the corresponding GOA file

Firstly, prior to GO enrichment analysis, OrthoGather verifies the availability of Gene Ontology Annotation (GOA) data for each analysed species and retains only those with corresponding annotations for downstream analysis (Fig. S9). Species for which a GOA file cannot be automatically matched can be manually corrected by providing a Taxon ID. If no valid GOA file is available, proteins from that species do not contribute GO annotations to the enrichment analysis, but they remain part of the orthogroup structure. Proteins without assigned GO terms may still be present but do not contribute to enrichment statistics.

GOA Species Matching Result			
Species to Analyze	Matched Entry in JSON	GOA Fi...	Fix
Acinetobacter_baumannii_(strain_ATCC_19606___DSM_30...	✓ <i>Acinetobacter baumannii</i> (strain ATCC 19606 / DSM 3...	✓	✓
Burkholderia_cenocepacia_(strain_ATCC_BAA-245___DSM...	✓ <i>Burkholderia cenocepacia</i> (strain ATCC BAA-245 / D...	✗	Taxon ID Fix
Escherichia_coli_(strain_K12)	✓ <i>Escherichia coli</i> (strain K12) [UP000000625]	✓	✓
Mycobacteroides_abscessus_(strain_ATCC_19977___DSM...	✓ <i>Mycobacteroides abscessus</i> (strain ATCC 19977 / DS...	✓	✓
Mycolicibacterium_smegmatis_(strain_ATCC_700084___m...	✓ <i>Mycolicibacterium smegmatis</i> (strain ATCC 700084 / ...	✓	✓
Pseudomonas_aeruginosa_(strain_ATCC_15692___DSM_2...	✓ <i>Pseudomonas aeruginosa</i> (strain ATCC 15692 / DSM ...	✓	✓
Download GOA Files			

Figure S9. Verification of GOA files availability. GOA files were automatically matched for five species. No GOA file was available in the GOA repository for *Burkholderia cenocepacia*; therefore, its proteins did not contribute GO annotations to the enrichment analysis, although the species remained included in the orthogroup structure.

After that, OrthoGather allows the evaluation of proportion of proteins that are associated with at least one GO term relative to the total number of proteins in the group. This provides a quantitative overview of GO annotation coverage across the dataset, which is visualised as summary distributions (Fig. S10). Across the selected species, the median proportion of annotated proteins per orthogroup is 66.67% when considering all orthogroups (Fig. S10 A), increasing to 70.52% when only orthogroups containing at least one annotated protein are included (Fig. S10 B). Fewer than 400 orthogroups lack functional annotation for all their constituent proteins (Fig. S10 C), whereas approximately 1,600 orthogroups are fully annotated, with functional information available for every protein in the group (Fig. S10 C & D). Overall, most orthogroups have annotation coverage between 50% and 70%.

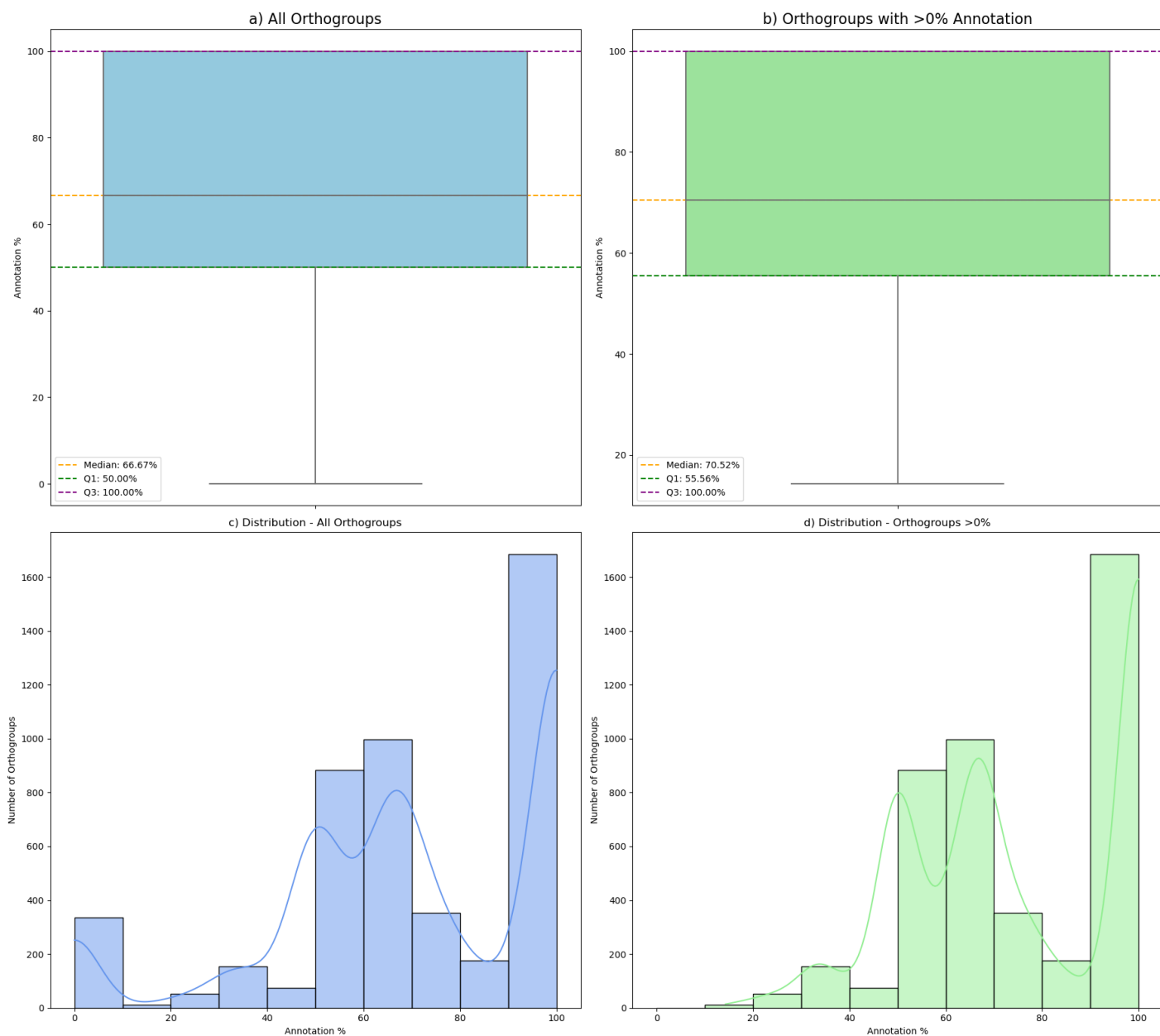


Figure S10. GO Annotation Distribution. Global annotation orthogroup coverage prior to Gene Ontology enrichment analysis. (a) and (b) Boxplots representing the percentage of annotated proteins per orthogroup; (a) complete set of orthogroups; (b) only orthogroups with at least one annotated protein. (c) and (d) Histograms illustrating the frequency distribution of annotation percentages across orthogroups; c) all orthogroups; d) only orthogroups with at least one annotated protein. A high quality figure is available in *S3. OrthoGather Outputs (GO_Annotation_Distribution.png)*

Further, in the enrichment phase, the user can perform a GO enrichment analysis for a defined subset of proteins, based on GO terms assigned to proteins within the selected orthogroups from species with available GOA files. To do that the user must define the foreground (proteins of interest) and background, to determine whether a GO term occurs more frequently in the foreground than expected from the background (Fig. S11). For this example analysis, we defined the foreground as the set of 596 proteins differentially expressed in *Msm* following rifampicin exposure and the *background* as the 3,244 proteins detected through mass spectrometry in the case of *Msm*.

Gene Ontology Enrichment Analysis

Foreground (UniProt IDs)

A0QR99
A0QZ46
A0R635
A0QQW4
A0R0C7

☒ Use Orthogroups

Foreground saved.

Background (Using UniProt IDs or pre-downloaded GOA files)

A0QR99
A0QZ46
A0R635
A0QQW4
A0R0C7

☒ Use Orthogroups ☐ Use all GOA files

Background saved. You can run the GO enrichment now.

Figure S11. Defining the Foreground and Background in OrthoGather Go term enrichment analysis module.

The *Use orthogroups* option was enabled, resulting in expansion of the foreground and background sets to include orthologous proteins across species, and enrichment was computed using their associated GO annotations. To determine whether a GO term occurs more frequently in the foreground than expected from the background, Fisher's exact test is applied. Multiple testing is performed using the Benjamini–Hochberg false discovery rate (FDR) correction. OrthoGather generates a figure with three separate bar plots displaying GO term enrichment for Biological Process, Molecular Function and Cellular Component (Fig. S6), and an accompanying data file submersing the statistics and results (S3. *Orthogather_outputs.zip* file, *GO_Enrichment_Report.xlsx*). The user can specify parameters of the analysis, including the FDR threshold, GO hierarchy depth, and maximum terms displayed (Fig. S12).

Adjust Parameters for Re-Generating Figure 9

Depth: P-value: Number of Terms:

Figure S12. Adjustment options for the Go term enrichment analysis

In this example dataset analysis, the false discovery rate (FDR) threshold was set to 0.001, the GO hierarchy depth to two, and a maximum of 50 terms were displayed. The top enriched Biological Process terms included signal transduction, alkanesulfonate metabolism, lipid metabolism, and antibiotic biosynthetic processes (Fig. S13; Table S5). Enriched Molecular Function terms were dominated by kinase-mediated phosphorelay signaling, oxidoreductase and monooxygenase activities, and ATP turnover-related functions such as ATP hydrolysis and ribonucleoside triphosphate phosphatase activity. The Cellular Component namespace

highlighted transporter-associated plasma membrane complexes, transcriptional and metabolic enzyme assemblies, and intracellular organelle-associated structures. Together, these enrichments indicate that exposure to sublethal concentrations of rifampicin induces a coordinated cellular response in *Msm*, consistent with known stress responses associated with antibiotic tolerance and resistance. Notably, the enrichment of RNA polymerase-associated complexes and kinase-mediated phosphorelay signaling aligns with rifampicin's primary inhibition of DNA-dependent RNA polymerase, suggesting compensatory reprogramming that may support survival. This analysis highlights how OrthoGather's orthogroup-focused enrichment framework leverages cross-species functional annotations to uncover biologically meaningful response signatures.

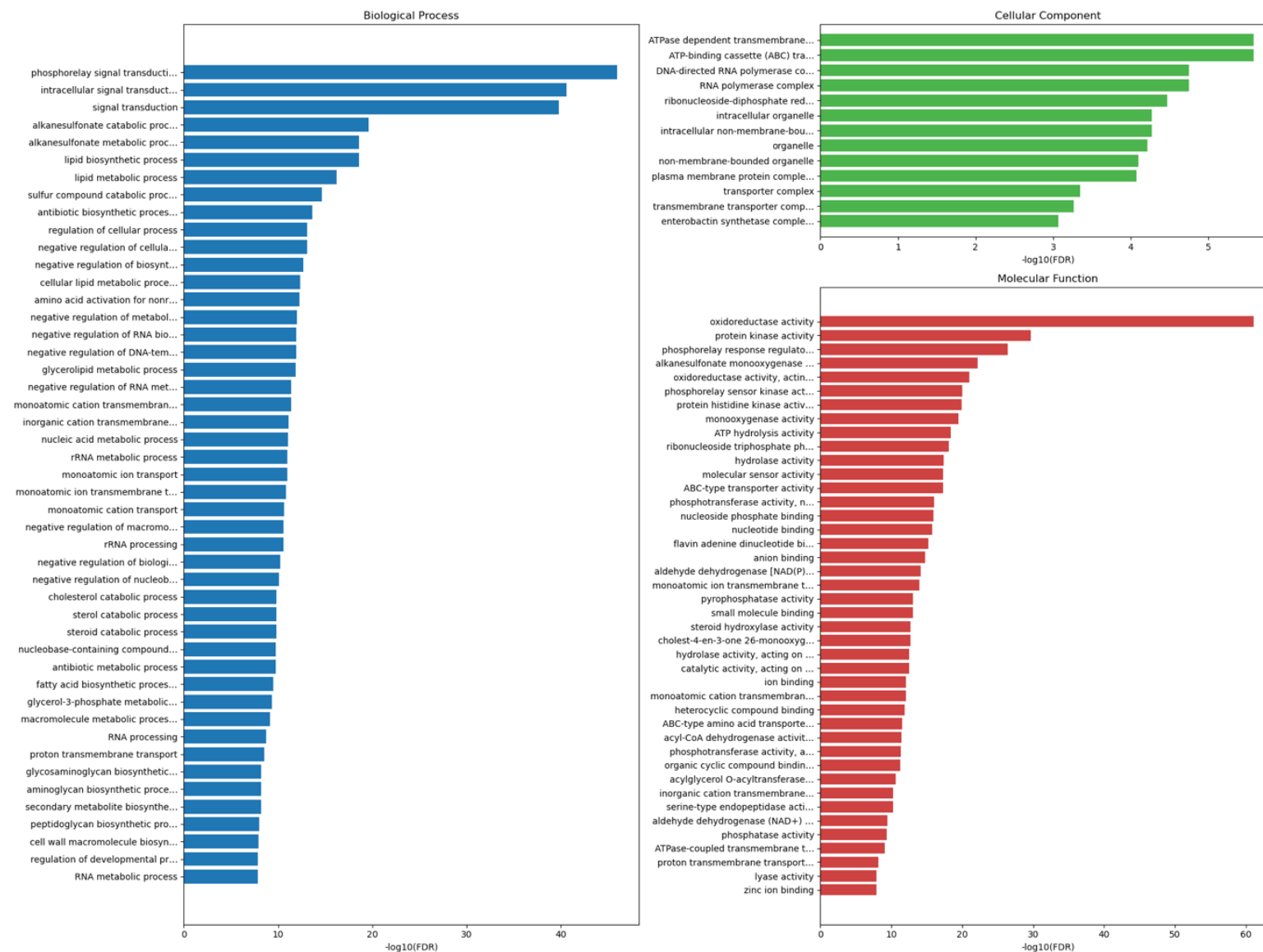


Figure S13. GO Enrichment Results. GO enrichment analysis using the foreground and background sets defined in *Supplementary Data S.1* The enrichment was performed using a significance threshold of $FDR < 0.001$, a maximum of 50 terms per category, and a GO hierarchy depth level of 2. The results are organized into the three main GO namespaces: Biological Process (BP) - blue bars; Molecular Function (MF) - red bars; Cellular Component (CC) - green bars. Each horizontal bar represents a GO term significantly overrepresented in the foreground relative to the background. The x-axis indicates the $-\log_{10}(\text{FDR})$ value, reflecting the statistical strength of enrichment, while the y-axis lists the GO term names. A high quality figure is available in *S3. OrthoGather Outputs (GO_Enrichment_Results.png)*

Table S5. Gene Ontology (GO) enrichment analysis of the example dataset (FDR < 0.001, GO hierarchy depth = 2, maximum 50 terms per ontology). The top ten enriched terms are shown for Biological Process and Molecular Function, and all significant terms are shown for Cellular Component. *GO*: unique identifier of the GO term; *Name*: name of the GO category; *NS*: ontology namespace; $-\log_{10}(FDR)$: negative logarithm of the adjusted p-value (higher values- stronger enrichment; *study_count*: number of annotated proteins in the foreground; *study_n*: total number of annotated proteins in the foreground; *pop_count*: number of annotated proteins in the background; *pop_n*: total number of annotated proteins in the background. The full output of this analysis can be found in *S3. OrthoGather Outputs (GO_Enrichment_Report.xlsx)*

GO	Name	NS	$-\log_{10}(FDR)$	study_count	study_n	pop_count	pop_n
Biological process							
GO:0000160	phosphorelay signal transduction system	biological_process	45.97	184	3390	245	11435
GO:0035556	intracellular signal transduction	biological_process	40.56	186	3390	262	11435
GO:0007165	signal transduction	biological_process	39.74	193	3390	278	11435
GO:0046306	alkanesulfonate catabolic process	biological_process	19.60	47	3390	49	11435
GO:0019694	alkanesulfonate metabolic process	biological_process	18.58	47	3390	50	11435
GO:0008610	lipid biosynthetic process	biological_process	18.57	225	3390	441	11435
GO:0006629	lipid metabolic process	biological_process	16.18	368	3390	843	11435
GO:0044273	sulphur compound catabolic process	biological_process	14.63	52	3390	64	11435
GO:0017000	antibiotic biosynthetic process	biological_process	13.59	47	3390	57	11435
GO:0050794	regulation of cellular process	biological_process	13.09	585	3390	1512	11435
Cellular component							
GO:0098533	ATPase dependent transmembrane transport complex	cellular_component	5.59	120	3390	266	11435
GO:0043190	ATP-binding cassette (ABC) transporter complex	cellular_component	5.59	120	3390	266	11435
GO:0000428	DNA-directed RNA polymerase complex	cellular_component	4.75	19	3390	24	11435
GO:0030880	RNA polymerase complex	cellular_component	4.75	19	3390	24	11435
GO:0005971	ribonucleoside-diphosphate reductase complex	cellular_component	4.47	11	3390	11	11435
GO:0043229	intracellular organelle	cellular_component	4.27	44	3390	260	11435
GO:0043232	intracellular non-membrane-bounded organelle	cellular_component	4.27	44	3390	260	11435
GO:0043226	organelle	cellular_component	4.22	45	3390	263	11435
GO:0043228	non-membrane-bounded organelle	cellular_component	4.10	45	3390	261	11435
GO:0098797	plasma membrane protein complex	cellular_component	4.07	122	3390	288	11435
GO:1990351	transporter complex	cellular_component	3.35	129	3390	318	11435
GO:1902495	transmembrane transporter complex	cellular_component	3.27	128	3390	316	11435
GO:0009366	enterobactin synthetase complex	cellular_component	3.07	8	3390	8	11435
Molecular Function							
GO:0016491	oxidoreductase activity	molecular_function	61.10	953	3390	2096	11435
GO:0004672	protein kinase activity	molecular_function	29.68	117	3390	153	11435
GO:0000156	phosphorelay response regulator activity	molecular_function	26.41	67	3390	72	11435
GO:0008726	alkanesulfonate monooxygenase activity	molecular_function	22.20	47	3390	47	11435
GO:0016614	oxidoreductase activity, acting on CH-OH group of donors	molecular_function	20.98	217	3390	406	11435
GO:0016705	oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	molecular_function	20.31	169	3390	294	11435
GO:0000155	phosphorelay sensor kinase activity	molecular_function	19.99	82	3390	108	11435
GO:0004673	protein histidine kinase activity	molecular_function	19.94	84	3390	112	11435
GO:0004497	monooxygenase activity	molecular_function	19.50	162	3390	282	11435
GO:0016887	ATP hydrolysis activity	molecular_function	18.40	239	3390	477	11435

